C++ Programming Basics

A curated collection of 10 essential introductory C++ programs with solutions

REFERENCE SHEET

01 Hello World & Basic Output

Demonstrates standard output printing using `std::cout` and including the basic `iostream` library.

```cpp
#include <iostream>

int main() {
    // Print greeting message to stdout
    std::cout << "Hello, World!" << std::endl;
    return 0;
}
```

SAMPLE OUTPUT:

```
Hello, World!
```

02 User Input and Sum of Two Numbers

Takes two integers from standard input (`std::cin`), calculates their sum, and displays the result.

```cpp
#include <iostream>

int main() {
    int a, b;
    std::cout << "Enter two integers: ";
    std::cin >> a >> b;
    int sum = a + b;
    std::cout << "Sum: " << sum << std::endl;
    return 0;
}
```

SAMPLE OUTPUT:

```
Enter two integers: 12 25
Sum: 37
```

03 Check Even or Odd Integer

Uses the modulus operator (%) inside an `if-else` conditional statement to test divisibility by 2.

```cpp
#include <iostream>

int main() {
    int num;
    std::cout << "Enter an integer: ";
    std::cin >> num;
    if (num % 2 == 0) {
        std::cout << num << " is Even." << std::endl;
    } else {
        std::cout << num << " is Odd." << std::endl;
    }
    return 0;
}
```

SAMPLE OUTPUT:

```
Enter an integer: 7
7 is Odd.
```

04 Find the Largest of Three Numbers

Demonstrates nested conditional logic and relational operators (>=, &&) to compare three values.

```cpp
#include <iostream>

int main() {
    double n1, n2, n3;
    std::cout << "Enter three numbers: ";
    std::cin >> n1 >> n2 >> n3;

    if (n1 >= n2 && n1 >= n3)
        std::cout << "Largest: " << n1 << std::endl;
    else if (n2 >= n1 && n2 >= n3)
        std::cout << "Largest: " << n2 << std::endl;
    else
        std::cout << "Largest: " << n3 << std::endl;

    return 0;
}
```

SAMPLE OUTPUT:

```
Enter three numbers: 14.5 9.2 27.8
Largest: 27.8
```

05 Factorial of a Number using Loop

Calculates $n! = 1 \times 2 \times \dots \times n$ using an iterative `for` loop and a 64-bit integer accumulator.

```cpp
#include <iostream>

int main() {
    int n;
    unsigned long long factorial = 1;
    std::cout << "Enter a positive integer: ";
    std::cin >> n;

    if (n < 0) {
        std::cout << "Factorial of negative numbers doesn't exist." << std::endl;
    } else {
        for (int i = 1; i <= n; ++i) {
            factorial *= i;
        }
        std::cout << "Factorial of " << n << " = " << factorial << std::endl;
    }
    return 0;
}
```

SAMPLE OUTPUT:

```
Enter a positive integer: 6
Factorial of 6 = 720
```

06 Fibonacci Series up to N Terms

Generates the Fibonacci sequence where each term is the sum of the preceding two terms ($0, 1, 1, 2, 3, 5, \dots$).

```cpp
#include <iostream>

int main() {
    int n, t1 = 0, t2 = 1, nextTerm = 0;
    std::cout << "Enter the number of terms: ";
    std::cin >> n;

    std::cout << "Fibonacci Series: ";
    for (int i = 1; i <= n; ++i) {
```

07 Reverse an Integer

Reverses digits of an integer using arithmetic operations (modulus % for last digit, division / to trim).

```cpp
#include <iostream>

int main() {
    int n, reversedNum = 0, remainder;
    std::cout << "Enter an integer: ";
    std::cin >> n;

    while (n != 0) {
        remainder = n % 10;
        reversedNum = reversedNum * 10 + remainder;
        n /= 10;
    }

    std::cout << "Reversed Number = " << reversedNum << std::endl;
    return 0;
}
```

SAMPLE OUTPUT:

```
Enter an integer: 12345
Reversed Number = 54321
```

08 Find Largest Element in an Array

Iterates through a contiguous static array to determine the maximum value present using linear search.

```cpp
#include <iostream>

int main() {
    int arr[] = {23, 45, 12, 89, 67, 34};
    int n = sizeof(arr) / sizeof(arr[0]);
    int maxVal = arr[0];

    for (int i = 1; i < n; ++i) {
        if (arr[i] > maxVal) {
            maxVal = arr[i];
        }
    }

    std::cout << "Array elements: 23, 45, 12, 89, 67, 34" << std::endl;
    std::cout << "Largest element: " << maxVal << std::endl;
    return 0;
}
```

SAMPLE OUTPUT:

```
Array elements: 23, 45, 12, 89, 67, 34
Largest element: 89
```

09 Check Prime Number

Determines whether a given positive integer is prime by testing divisibility up to $\sqrt{n}$.

```cpp
#include <iostream>

int main() {
    int n;
    bool isPrime = true;
    std::cout << "Enter a positive integer: ";
    std::cin >> n;

    if (n <= 1) isPrime = false;
    else {
```

10 Check Palindrome String

Uses standard `std::string` and a two-pointer method to check if a word reads the same forwards and backwards.

```cpp
#include <iostream>
#include <string>

int main() {
    std::string str;
    std::cout << "Enter a string: ";
    std::cin >> str;

    int left = 0;
    int right = str.length() - 1;
    bool isPalindrome = true;

    while (left < right) {
        if (str[left] != str[right]) {
            isPalindrome = false;
            break;
        }
        left++;
        right--;
    }

    if (isPalindrome)
        std::cout << "\"" << str << "\" is a Palindrome." << std::endl;
    else
        std::cout << "\"" << str << "\" is NOT a Palindrome." << std::endl;

    return 0;
}
```

```
SAMPLE OUTPUT:
Enter a string: radar
"radar" is a Palindrome.
```

Summary of Covered C++ Core Concepts

Program #	Core Concept	Key Operators / Standard Library
1 – 2	I/O Streams & Variable Types	`std::cout`, `std::cin`, `<iostream>`
3 – 4	Conditionals & Decision Making	`if-else`, Modulus %, Logical AND &&
5 – 7	Iteration & Loops	`for` loops, `while` loops, accumulator pattern
8 – 10	Arrays, Strings & Algorithms	Static arrays, `std::string`, two-pointer technique